

Counterfactual Explanations of AI Systems

Fundamentals, Methods, & User Studies for XAI

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The Problem

ChatGPT OpenAI



<https://youtu.be/eMx-2s7mZ24>

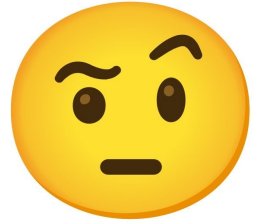


AI is out there...being deployed...busily making mistakes !

Spot the Ball !



Spot the Head !



One Solution: Explain It !

Ante-Hoc / intrinsic:

Design the model so that its decision process is interpretable by construction.

Post-Hoc: Generate an explanation* for the behavior of an already trained model.

- understand individual predictions
- diagnose errors
- audit systematic model behavior

* BEWARE: An explanation is not automatically a faithful account of the model's internal computation.



Post Hoc: Making a Misclassification Less Mysterious

Test Image



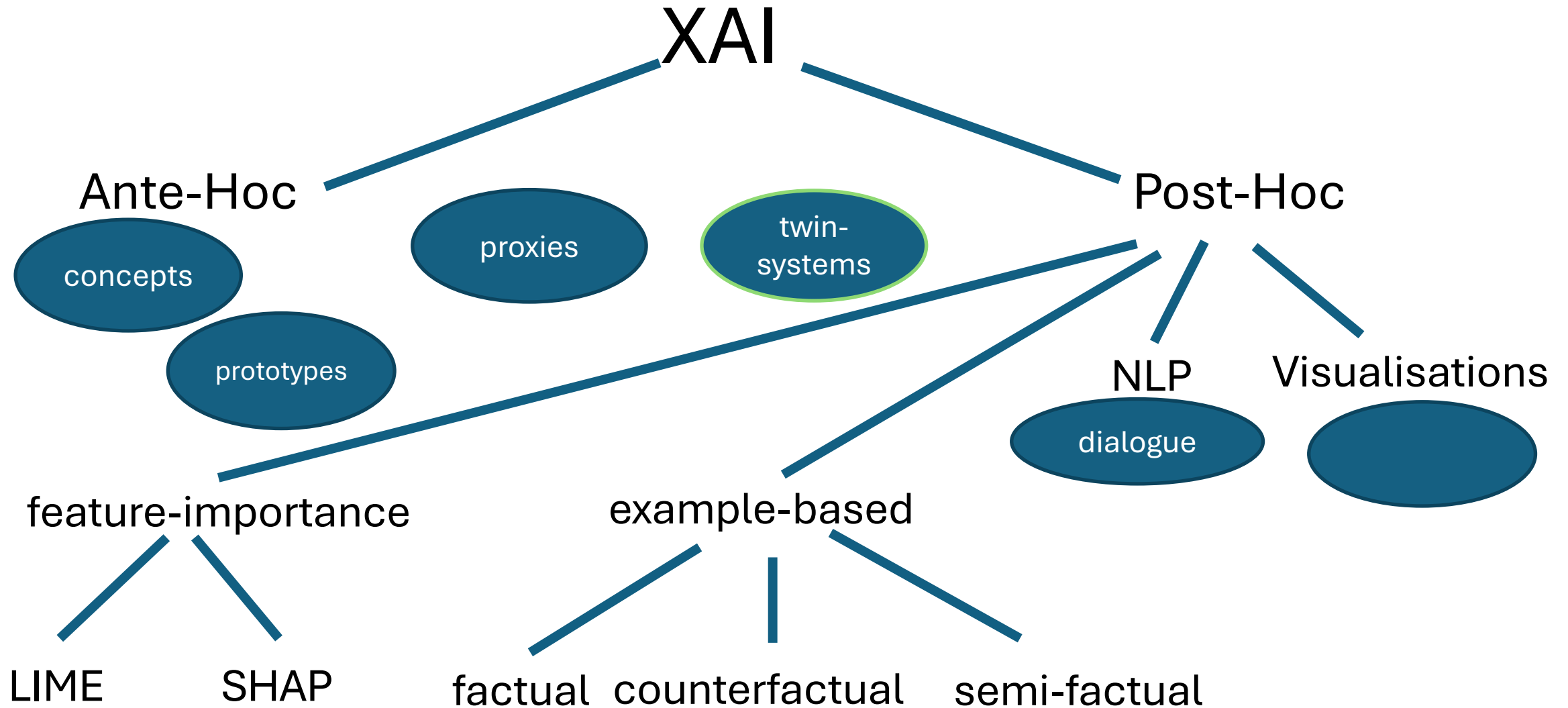
Label: **Rifle**
Classification: **Shovel**

Explanation-by-Example

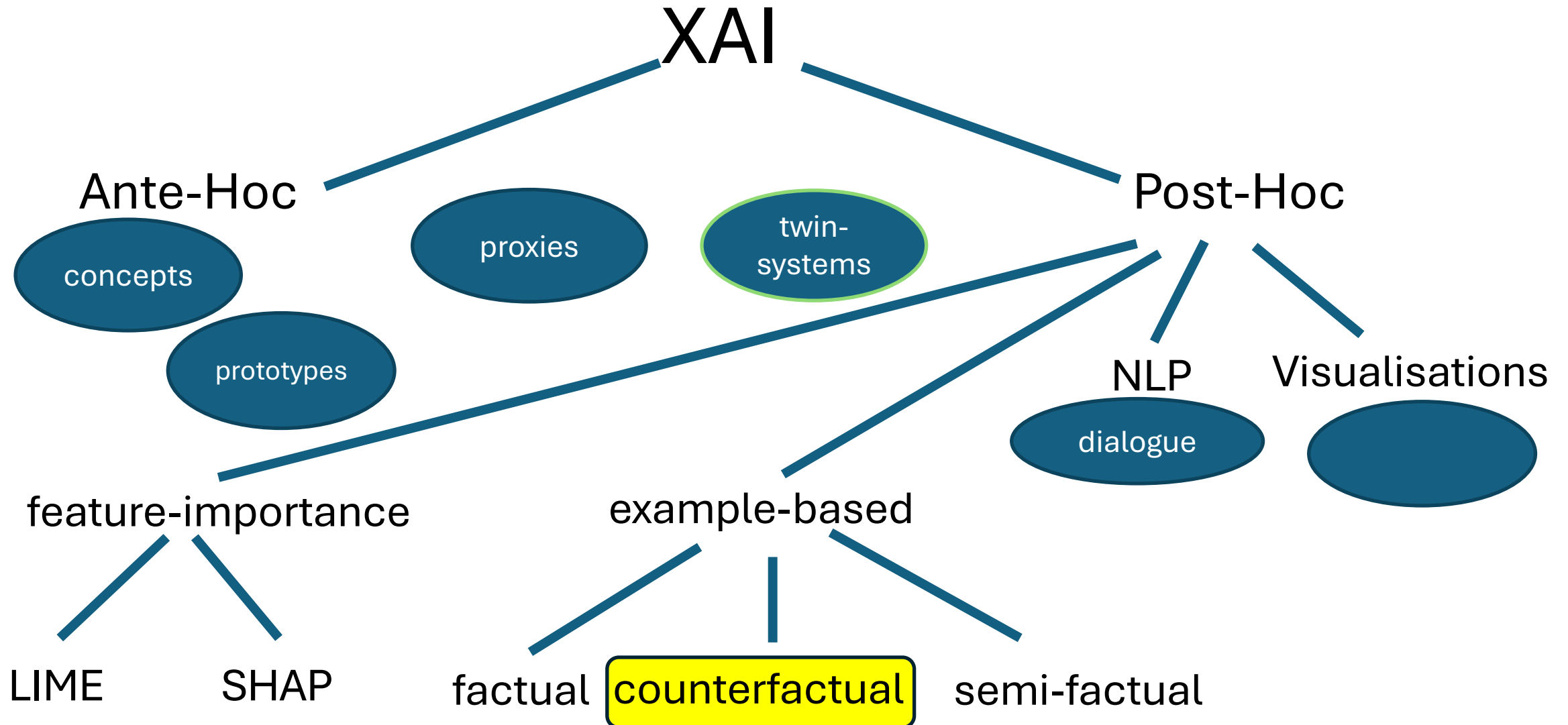


Label: **Shovel**
Classification: **Shovel**

Explainable AI: Taxonomy



Explainable AI: Taxonomy



Counterfactual Explanations

***Your loan
application has
been refused !***

***If you earned 5k
more, you would have
gotten the loan.***

XAI



Counterfactual Explanations

If the test image looked like this, I would have thought it was an “8”.



Test Image

Label: 8
Prediction: 3

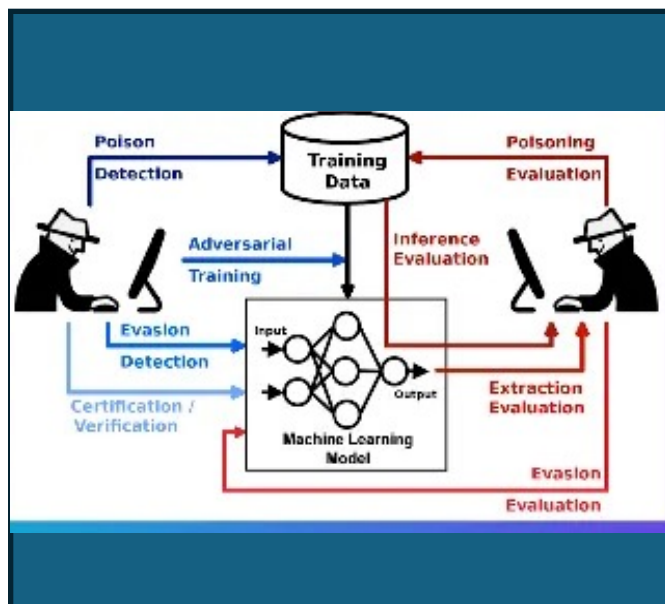


Counterfactual

New Prediction: 8

Why Counterfactuals?

Artificial Intelligence



Looks like adversarial learning...

[Wachter et al \(2019\) Facct-19](#)

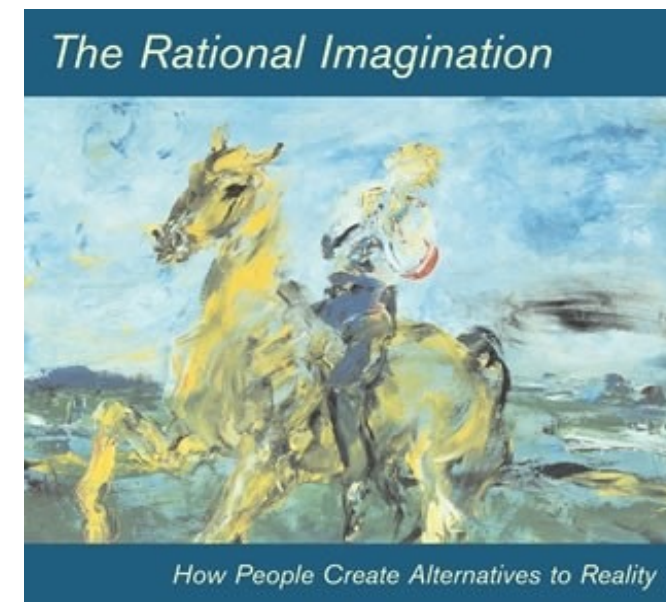
Legal



GDPR compliant...

[Wachter et al \(2017\) Harv. JL &B Tech.](#)

Psychology



Used by people spontaneously...

[Byrne \(2007\) MIT Press](#)

Plan for the Day...

Time	Topics
1:30 PM	Introduction
	<i>Hello and Introductions!</i>
1:45 PM	From Philosophy of Counterfactuals to XAI and User Studies
	<i>Philosophical & Psychological Fundamentals of Counterfactuals</i>
	<i>Getting Into the User Study Mindset</i>
	Q&A
2:20 PM	Computational Fundamentals of Counterfactuals in AI
	<i>Formalisation</i>
	<i>Current and Future Directions</i>
	Q&A
3:00 PM	COFFEE (15:00h-15:30h)
3:30 PM	Hands-On I: Using a Counterfactual Toolbox
	<i>From a Simple Baseline to Diverse Counterfactuals</i>
	Q&A
4:10 PM	Fundamentals of XAI User Studies
	<i>A Roadmap: From Motivation to Results</i>
	Q&A
4:50 PM	Hands-On II: User Sample Study - Walkthrough and Analysis
	<i>From Raw Data to a Clean Analysis Sample</i>
	<i>Testing & Interpreting Effects of Counterfactual Explanations</i>
	Q&A
5:20 PM	Closing Session, Discussion and Final Q&A
TUTORIAL END	

